# Product Family Design - Variant

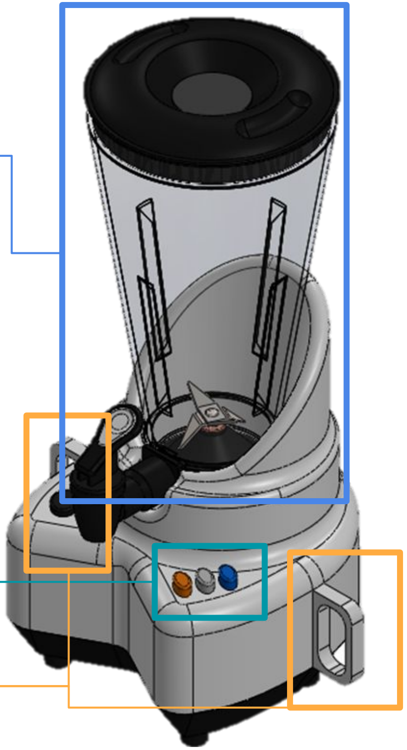
## Smoothie maker for Household

## Smoothie maker for Caterer

**Module change I: Container**
Increase jar volume from 2L to 3L for larger capacity

**Module change II: Color-differentiated buttons**
More straightforward to use, improve operator speed

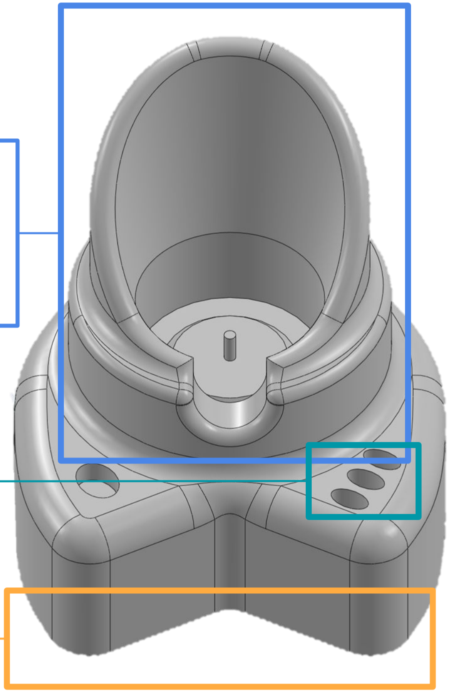
**Module change III: Main body with handles**
Added handles on the main body for easier transporting during events



# Product Family Design - Common Interfaces

## Smoothie maker for Household

## Smoothie maker for Caterer



**Interface I: Jar Mount & Drive Coupling**

Same bayonet geometry, O-ring seat, and motor-to-blade coupling across variants

**Interface III: Control Buttons**

**Interface II: Platform Mounting**

Same base (foot/boss pattern, datum faces)

# Design for Manufacturing - Metal

## Post DFA, Pre DFM

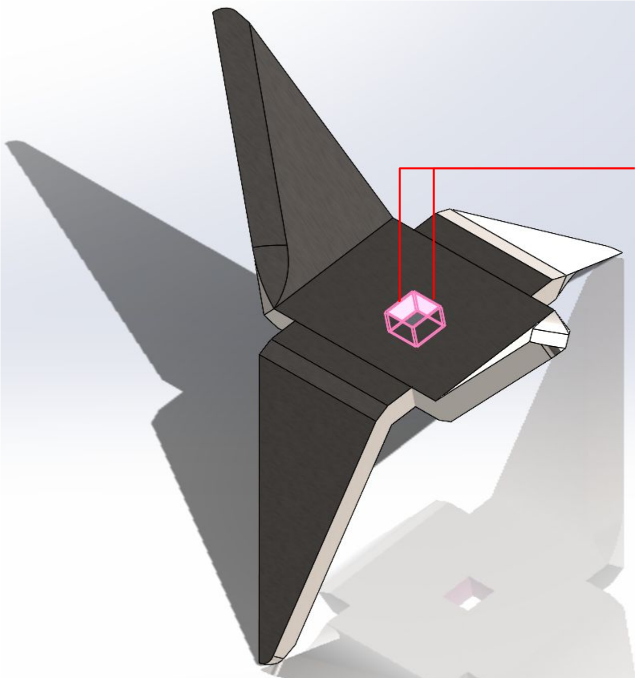


Fig. DFMPro Recommendation

DFMPro recommended increasing the “slot width to thickness ratio” to 2 or more (Use Slot Width >= 4.0 mm) (Currently ratio is 3 mm / 2 mm = 1.5)

**DFSM-7: avoid narrow features**

Material: AISI 304 stainless steel
Thickness: 2 mm

| Cost of Sheet Metalworking per part | |
|---|---|
| Material Cost | $ 0.2380 |
| Operating Cost | $ 0.1634 |
| Die Cost* | $ 0.0003 |
| **Total cost per part** | **$ 0.4017** |

*Blanking, Bending and Piercing dies required

**Rule Description**

If the slot width falls below a minimum value, compressive stress on the punch increases dramatically, causing the thin and weak punch to buckle or break. Hence the slot width should increase with increasing slot length and sheet thickness.

# Design for Manufacturing - Metal

MAKE THE WORLD WORK BETTER

## Post DFA and DFM

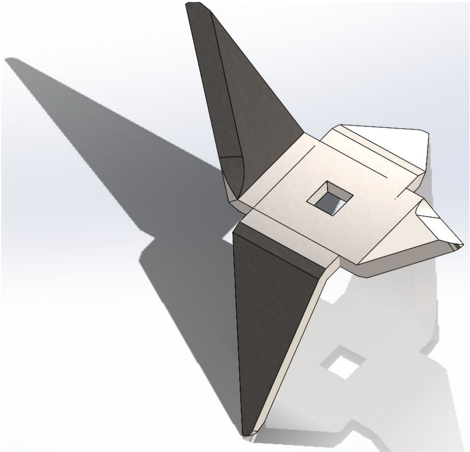


Fig. Post DFM part isometric view

**DFSM-7: avoid narrow features**

Changed slot width from 3.0 mm to 4.0 mm

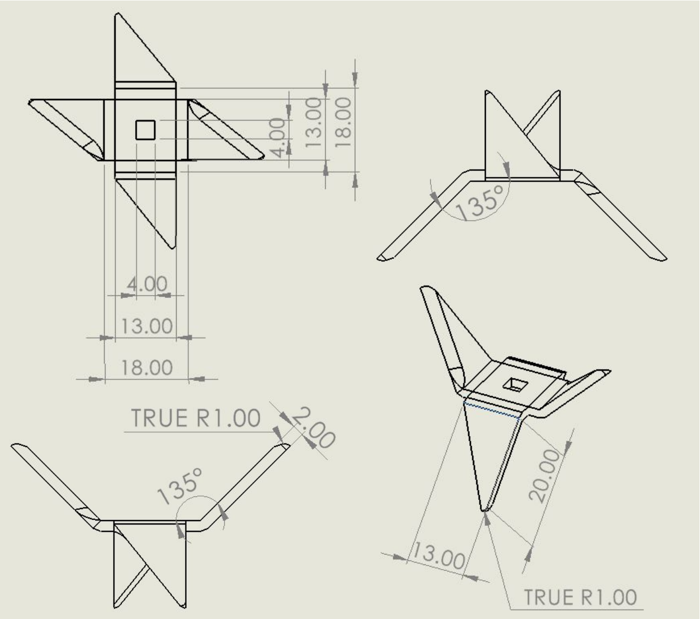



Fig. Additional Views for Post DFM part

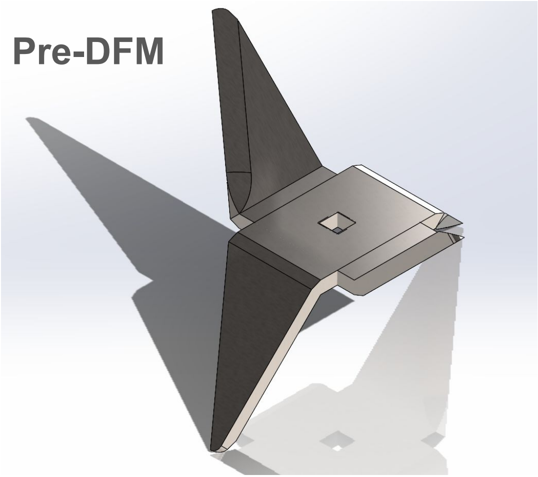
| Cost of Sheet Metalworking per part | |
|---|---|
| Material Cost | $ 0.2380 |
| Operating Cost | $ 0.1636 |
| Die Cost* | $ 0.0003 |
| **Total cost per part** | **$ 0.4019** |

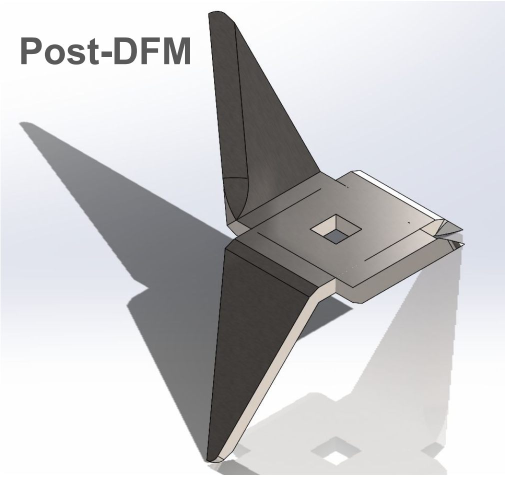
*Blanking, Bending and Piercing dies required

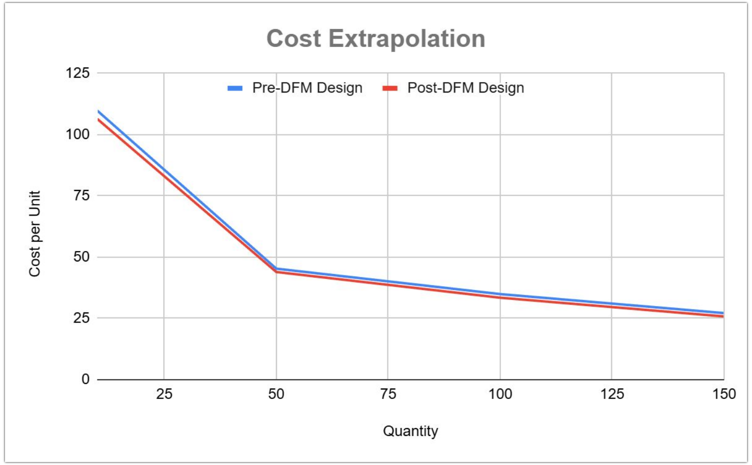
**Increased cost per part by $ 0.0002**

# Design for Manufacturing - Metal

## Cost Comparison




| | Analytical Method* | Hubs** |
|---|---|---|
| Pre-DFM Design | $ 0.4017 | $ 54.31 |
| Post-DFM Design | **$ 0.4019 (higher)** | **$ 52.41 (lower)** |

*Analytical method used **sheet metalworking process**
**Hubs quote used **machining process** due to lack of quote availability for sheet metalworking